OpenShift Persistent Storage

This page is to document the details of OpenShift Container Platform Persistent storage and how to use it.

Overview	
Scope	OpenShift Container Platform GA
Status	DEV
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OpenShift Data Foundation

To support persistent data workloads in OpenShift Container Platform, we deploy a Redhat Operator called OpenShift Data Foundation (ODF).

Architecture

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We deploy ODF on dedicated storage nodes in each cluster locally. Our current deployment is done using VMWare disks. To ensure data redundancy we have 3 replicas for each volume spread across different storage nodes.

Storage Classes

We primarily support using the below storage classes, deployed by ODF. Both storage options are dynamically provisioned, meaning you do not require intervention from the Container Platforms administrators. You will be bound by storage quotas, however.

Storage Class	Access Modes	Storage Type	Volume Expansion Supported
ocs-storagecluster-ceph-filesystem	ReadWriteOnce (RWO) ReadWriteMany (RWX)	Filesystem	true
ocs-storagecluster-ceph-rbd	ReadWriteOnce (RWO) ReadWriteMany (RWX)	Raw Block Device	true

Disaster Recovery

OpenShift Data Foundation uses an OpenShift Operator called OADP (OpenShift APIs for Data Protection). This operator uses Velero Resources under the hood.

Please see OpenShift Backup and Restore for more information.